



2026 BASIC ABSTRACT ALGEBRA STUDY

LECTURE 1 HOMEWORKS

SOLUTION - Q9

Version

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Read this carefully before moving on.

Please, clearly specify whether the statement in question is *true* or *false*, for the *prove* or *disprove* type of questions. As always, it is highly encouraged for you to write each step clearly with full sentences.

As a demonstration of a proof answer, consider the following example.

The above statement is **true**.

Proof. ...

□

As a demonstration of a disproof answer, consider the following example.

The above statement is **false**.

Proof. ...

□

Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$ for an integer n . i.e., the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B , A is said to be a subset of B if the following condition holds.

If x is an element of A , then x is an element of B .

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A . This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.

EASY

Let A and B be two sets. Let $A(x)$ mean x is an element of A , and $B(x)$ mean x is an element of B .

Which of the following is equivalent to saying that $A \subset B$?

- ☐ 1. $B(x) \rightarrow A(x)$
☐ 2. $\text{not } A(x) \rightarrow B(x)$
☐ 3. $\text{not } A(x) \rightarrow \text{not } B(x)$
- ☐ 4. $A(x) \rightarrow \text{not } B(x)$
☐ 5. $\text{not } B(x) \rightarrow \text{not } A(x)$

Definition 4 (Roots of unity)

Let μ_n denote the set of all complex solutions to an equation of form $x^n = 1$ for a positive integer n . Elements of such a set are called n -th roots of unity.

For instance, $\mu_2 = \{1, -1\}$.

Q2.

EASY

Show that $\mu_2 \subset \mu_4$.

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m .

Hint 5. Provide concrete n and m that make up a counterexample. Try justifying your answer by using the definition of set builder notation.

Q4.

HARD

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 6. cf. lecture note.

Q5.

HARD

Show using Definition 2 that for any two positive integers n and m , $\mu_n \subset \mu_{nm}$.

Hint 7. Of course, it is a generalization of Q2.

Q6.

HARD

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n .

Hint 8. cf. PCSA Q4.

HARD

Q7.

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n , then either $n = 1$ or $n = -1$.

Hint 9. What are the factors of 1 and -1 ?

Definition 10 (Symmetric difference)

$A \triangle B$ is a binary set operation given by $(A \setminus B) \cup (B \setminus A)$ and is called a *symmetric difference* between A and B .

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that $A \triangle B \subset C$ if and only if $A \cup B \subset (A \cap B) \cup C$.

Q9.

LEVI

For an integer n , let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P be the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

(a) Show that every two distinct elements of P are disjoint.

Hint 11. Use Definition 3 and *contraposition*.

Proof. Let us rephrase the statement first. The original statement is $M(a) \neq M(b) \rightarrow M(a) \cap M(b) = \emptyset$ for all $a, b \in \mathbb{Z}$. Its contrapositive is $M(a) \cap M(b) \neq \emptyset \rightarrow M(a) = M(b)$ for all $a, b \in \mathbb{Z}$.

Assume $M(a) \cap M(b) \neq \emptyset$ for some $a, b \in \mathbb{Z}$. As it is not empty, there exists at least one element c in $M(a) \cap M(b)$. By the definition, $c \equiv a \pmod{n}$ and $c \equiv b \pmod{n}$.

If x is an element of $M(a)$, then $x \equiv a \equiv c \equiv b \pmod{n}$ by the symmetry and transitivity of the congruence. Hence, x is an element of $M(b)$. Similarly, if x is an element of $M(b)$, then $x \equiv b \equiv c \equiv a \pmod{n}$, showing that $M(a) = M(b)$ as desired. $\square$

If you're up for a challenge, try showing the symmetry and transitivity of the congruence used here.

The real name of $M(k)$ is *coset*. You just proved that they create a complete *partition* on $\mathbb{Z}$!

(b) Show that the following holds for any two integers a and b by Definition 3.

$$M(a) + M(b) = M(a + b)$$

Proof. Let a and b be integers.

Forward. If x is an element of $M(a) + M(b)$, there exist two integers y, z such that $y \equiv a \pmod{n}$ and $z \equiv b \pmod{n}$ and $x = y + z$. As $x - a - b = (y - a) + (z - b)$ is a multiple of n , $x \in M(a + b)$.

Backward. If x is an element of $M(a + b)$, then $x = k + a + b$ where k is a multiple of n . Then, there exist two integers $k + a, b$ such that $k + a \equiv a \pmod{n}$ and $b \equiv b \pmod{n}$ and $x = (k + a) + b$.

By extensionality, we conclude that $M(a) + M(b) = M(a + b)$. $\square$

In such a case, we often say that *the operation is well-defined on cosets*. The subgroups for which this coset operation is well-defined are exactly the *normal* ones; in an abelian group such as $\mathbb{Z}$, every subgroup, including $n\mathbb{Z}$, is normal. Moreover, we can now view M as a special kind of function, called a *canonical homomorphism to the quotient*.

Q10.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2^{121}} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: _____

Q11.

LEVI

Let T be the set $\{z \in \mathbb{C} \mid |z| = 1\}$ and μ_∞ be the set of all roots of unity as follows.

$$\mu_\infty = \bigcup_{n=1}^{\infty} \mu_n = \mu_1 \cup \mu_2 \cup \dots$$

Prove or disprove that $T = \mu_\infty$.

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark $\times$ in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark $?$ in the Evaluation column.
- If you understood the question and are sure you have solved it, mark $\circ$ in the Evaluation column.

No.	Evaluation	No.	Evaluation
Q1		Q7	
Q2		Q8	
Q3		Q9	
Q4		Q10	
Q5		Q11	
Q6			

There are these additional questions as well; feel free to answer them.

- (a) Which was the hardest for you, among those you have solved? _____
- (b) Which seemed to be the hardest for you, among those you haven't solved? _____
- (c) Which was your favorite question, if you had to choose one? _____